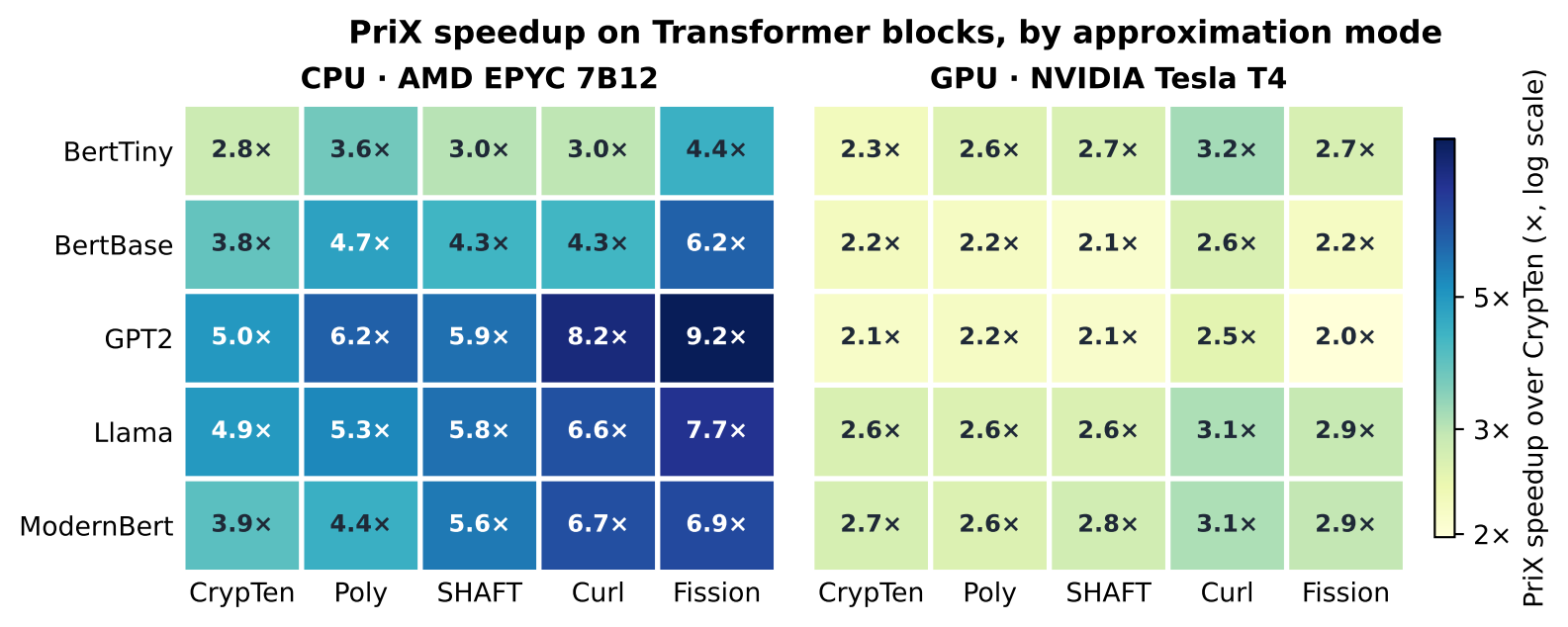
PriX speedup on Transformer blocks, by approximation mode
CPU · AMD EPYC 7B12
GPU · NVIDIA Tesla T4
BertTiny
BertBase
GPT2
Llama
ModernBert
2.8× 3.6× 3.0× 3.0× 4.4×
3.8× 4.7× 4.3× 4.3× 6.2×
5.0× 6.2× 5.9× 8.2× 9.2×
4.9× 5.3× 5.8× 6.6× 7.7×
3.9× 4.4× 5.6× 6.7× 6.9×
2.3× 2.6× 2.7× 3.2× 2.7×
2.2× 2.2× 2.1× 2.6× 2.2×
2.1× 2.2× 2.1× 2.5× 2.0×
2.6× 2.6× 2.6× 3.1× 2.9×
2.7× 2.6× 2.8× 3.1× 2.9×
CrypTen Poly SHAFT Curl Fission
CrypTen Poly SHAFT Curl Fission
PriX speedup over CrypTen (×, log scale)
5×
3×
2×